

Berk Gökden

Senior AI/ML Engineer · I build and run production AI systems end to end: agents, LLM platforms, data · Klarna · Booking.com · ING

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Professional Summary

Senior, hands-on AI/ML engineer with 15+ years in production software. I design, build and run AI systems end to end, from the first prototype to a live system that people use every day. Independent consultant since 2021 for Klarna, Booking.com, ING and others, now through my own company, PrimAxiom. At Klarna I built one of the company's internal AI platforms, used by 5,000+ employees. I work directly with product and business stakeholders, mentor engineers, and earlier led a 12-person engineering team at Vamp.io (acquired by CircleCI). Specialties: AI agents and generative AI (GenAI) applications, RAG, LLM fine-tuning, evaluation and LLMOps, and the data and infrastructure underneath. I also know the stack below the API: I wrote a custom sampler for llama.cpp and publish my own multilingual models on Hugging Face.

Highlights

- Built one of Klarna's internal AI platforms, used by **5,000+ employees**, with agents for meetings, voice, feedback and search — meeting efficiency up **~40%**, lower cost per meeting
- **Design, build, and deploy production AI agent systems** end-to-end — multi-agent orchestration, MCP tools, evaluation, and guardrails, from prototype to live systems
- Cut infrastructure cost **90%+** with an AI load-prediction optimizer at Vamp.io (acquired by CircleCI)
- **15+ years of production software engineering** — distributed systems, microservices, and Kubernetes across Go, Python, and TypeScript
- Build and run my own open-source AI systems: **ReasonGraph** (memory for AI agents, also run as a hosted service) and **Assay** (decision models from 149M to 27B parameters), plus multilingual models on Hugging Face

Professional Experience

Machine Learning Engineer (Consultant via PrimAxiom) 2026 - Present
Booking.com

- Amsterdam, Netherlands
- Build AI agents for the Booking.com AI Trip Planner and Booking Holdings' connected-trip strategy
 - Run agent and LLM evaluation and prompt optimization to improve accuracy, latency, and cost
 - Integrate MCP (Model Context Protocol) for tool and data connectivity
 - Technologies: Python, OpenAI, fine-tuned open-weight LLMs, Arize, Snowflake, MCP

Founder and Director 2026 - Present
PrimAxiom Labs

- Amsterdam, Netherlands
- My own Dutch company (primaxiom.ai), through which I consult for clients (currently Booking.com); I also build and run small software and AI products
 - Recent work: small language models that run on CPU, and agents with memory; the code and models are public on GitHub and Hugging Face
 - Technologies: Python, Go, TypeScript, LangGraph, Claude Agent SDK, vLLM, llama.cpp, PostgreSQL+pgvector, Kubernetes, AWS, GCP

Senior AI Engineer (Contract)

Sep 2023 - Dec 2025

Klarna

Amsterdam, Netherlands

- Built one of Klarna's internal AI platforms, used by 5,000+ employees, with agents for meetings, voice, feedback and search
- Built a voice meeting-intelligence agent — speech-to-text transcription (self-hosted Whisper first, later ElevenLabs) with LLM summarization and structured extraction of decisions and action items — reducing cost per meeting and improving meeting efficiency (~40%); also used beyond internal meetings, in meetings with Klarna's partners and in interviews
- Deployed GenAI applications using LangChain with OpenAI and Anthropic LLMs under enterprise agreements with custom RAG implementations
- Conducted systematic evaluation of LLM performance across different models (OpenAI, Anthropic, Hugging Face models such as Llama, Mistral) and parameters, iterating on prompts, RAG configurations, and model selection to optimize for accuracy, latency, and cost
- Fine-tuned LLMs and embedding models from Hugging Face for domain-specific terminology and internal knowledge bases
- Implemented vector search using PostgreSQL+pgvector, FAISS, and Annoy for cost-effective semantic search
- Built and deployed Slack-integrated AI agents connected to internal workflows and a RAG knowledge base — employees query company knowledge and trigger actions conversationally, in the tool they already use
- Deployed multi-agent systems and automated workflows that saved teams hours of manual tracking each week
- Shipped these systems via CI/CD to a containerized internal platform on AWS
- Technologies: Python, TypeScript, Go, LangChain, OpenAI (via Azure OpenAI), Anthropic, Hugging Face, Whisper, ElevenLabs, RAG, PostgreSQL+pgvector, FAISS, Docker, CI/CD, AWS

Senior Software Engineer - Data (Contract)

May 2023 - Sep 2023

Booking.com

Amsterdam, Netherlands

- Supported cloud migration from on-premise to AWS, facilitating data quality during infrastructure modernization
- Developed data validation services and pipelines using microservices architecture
- Improved data quality through a governance framework, meeting Booking's 99% pipeline-reliability requirement
- Validated millions of booking records daily across large-scale data processing systems
- Technologies: DBT, Apache Spark SQL, Snowflake, Hadoop, Java, Python, Argo, Kubernetes, AWS

Senior Data Engineer (Contract)

Jul 2022 - Apr 2023

ING

Amsterdam, Netherlands

- Automated environmental sustainability data processing for ING Wholesale Banking, reducing processing time from weeks to hours
- Built reliable daily pipeline tracking sustainability metrics for hundreds of companies used in credit score calculations, replacing manual Excel workflows
- Developed framework and tools to help data engineers build production pipelines faster
- Solved complex entity resolution challenges tracking company histories through mergers, closures, and restructuring
- Technologies: Python, Apache Spark, PySpark, Kedro, Docker, Airflow, Kubernetes (OpenShift)

Big Data Engineer / ML Engineer (Freelance)

Sep 2021 - Feb 2022

VodafoneZiggo

Utrecht, Netherlands

- Built predictive ML models to detect TV/internet box failures before they occurred, analyzing heat patterns and device telemetry
- Automated proactive customer support by triggering service calls when boxes showed failure patterns, reducing box-related complaints by ~70%, measured through call-center metrics
- Developed forecasting models to predict volume of failing boxes for capacity planning and inventory management
- Delivered ML pipelines to production in collaboration with data scientists
- Technologies: AWS (S3, EMR, Glue, Athena, DynamoDB, Lambda), Python

Machine Learning Engineer

Sep 2020 - Aug 2021

DPG Media Netherlands

Amsterdam, Netherlands

- Built search re-ranking system for DPG Media magazine app using user history while avoiding filter bubbles
- Developed notification optimization models tracking user engagement patterns to improve message interactions
- Built user data pipeline merging clickstream data from multiple websites and apps across DPG Media acquisitions to create unified user profiles for ad targeting
- Implemented ML models including classification, clustering, and decision trees for user segmentation, personalization, and content recommendations
- Won company hackathon by building vector-based recommendation system using Universal Sentence Encoder for article enrichment and fast search, first transformer-based system deployed at DPG Media
- Developed streaming applications using Apache Kafka (AWS MSK) and Spark Streaming for real-time data processing
- Technologies: Python, scikit-learn, pandas, numpy, Apache Kafka, Spark, Delta Lake, AWS (S3, DynamoDB, EKS), Airflow, MLflow, Databricks (evaluated), FastAPI, Flask, Terraform, Universal Sentence Encoder

Senior Software Engineer

Caspar AI

Mar 2020 - Sep 2020

Rotterdam, Netherlands

- Built AI sensor system for elderly care facilities in US, monitoring falls, controlling lighting/heating, and alerting nurses
- Set up on-premise Kubernetes clusters inside buildings for privacy compliance, ensuring no user data left facilities
- Deployed deep learning models for image recognition on edge servers, processing data locally for security
- Built data pipeline sending only operational metrics to cloud while keeping sensitive data on-premise
- Technologies: Python, Kubernetes (Kubespray), TensorFlow Serving, Nginx, AWS (S3, EC2, MSK), Kafka

Engineering Lead

Vamp.io (acquired by CircleCI)

Oct 2017 - Mar 2020

Amsterdam, Netherlands

- Led engineering team of 12 people building cloud-native AIOps platform for canary releasing and microservices deployment
- Built distributed systems and container orchestration infrastructure on Kubernetes
- Developed AI load prediction algorithm for Vamp Continuous Cloud Optimizer, optimizing CPU and memory usage and reducing infrastructure costs by over 90%
- Company subsequently acquired by CircleCI
- Technologies: Go, Kubernetes, Docker, microservices architecture, machine learning

Senior Software Engineer

Zoover / Weeronline

Sep 2016 - Sep 2017

Amsterdam, Netherlands

- Rebuilt weather data ETL pipeline for Weeronline app/website, reducing radar image and rain prediction latency from 45 minutes to under 1 minute
- Rapidly built fraud detection system using Apache Spark analyzing review patterns and user behavior to identify and flag fraudulent reviews on Zoover platform
- Set up CI/CD pipelines with Kubernetes (GKE), CircleCI, and GitLab using Scala

Developer

SAP SE

Mar 2015 - Sep 2016

Istanbul, Turkey / Walldorf, Germany

- Developed SAP Vora in-memory query engine integrated with Apache Spark for Big Data analytics on Hadoop
- Built Docker-based deployment system for running Spark clusters in cloud environments
- Worked on database internals and cloud service integrations for SAP Vora

Software Engineer

T2 Software

Nov 2013 - Mar 2015

Ankara, Turkey

- Developed full-stack solutions for major Turkish telecom operator handling 100M+ subscriber records
- Built customer service systems including provider transfer functionality completing migrations in under 10 minutes
- Built service-oriented architecture and improved backend performance
- Negotiated with stakeholders to improve API design
- Technologies: Java, JavaScript, Oracle SQL

Software Engineer

Aselsan

Sep 2011 - Nov 2013

Ankara, Turkey

- Developed real-time simulation system for radar test laboratory testing missile guidance systems
- Built flight scenario simulator commanding radar hardware while logging and visualizing test results
- Implemented distributed architecture using multi-threading and reflective memory in C/C++ on Linux

Open Source Contributions & Projects

A few open-source pieces that show how I work below the API.

ReasonGraph — Memory for AI Agents

github.com/bgokden/reasongraph

Graph memory for AI agents: stores facts, extracts entities and cause → effect relations with small open models, and answers "why" questions with a cited chain of facts. Python library on PyPI, MCP server, and a hosted service I run (memory.primaxiom.ai). Measured in six languages in a public benchmark (reasongraph-bench), including the experiments that did not help.

Assay — Calibrated Decisions from a Language Model

github.com/bgokden/assay

Answers many typed questions about a text in one forward pass of a language model, without generating text, and returns calibrated probabilities plus whether the text contains enough evidence to answer. Includes the full training pipeline (data, training, calibration, evaluation); six models from 149M to 27B published on Hugging Face.

Small mDeBERTa models published on Hugging Face with ONNX versions for fast CPU inference: place extraction in 13 languages, and a multilingual cause/effect tagger (causal-span-model) used by ReasonGraph. Related tools: synthspan (synthetic NER training-data generator) and place-extractor-mcp (MCP server that gives AI assistants these models).

llama-constrain — Constrained LLM Generation

github.com/bgokden/llama-constrain

Developed custom sampler in llama.cpp enabling strict constrained sampling in LLMs. Library for constrained text generation with Large Language Models for structured output using GGUF models.

Annoy — Vector Search Library (Contributor)

github.com/spotify/annoy

Contributor to Spotify's approximate nearest neighbors library for vector search. Built custom implementations for production use cases.

Technical Skills

AI/ML

- **Agents:** Multi-agent orchestration, agent memory (short-term working + long-term graph and vector-backed), RAG, MCP, tool use, LangChain/LangGraph, Claude Agent SDK, Claude Code & coding agents
 - **Eval & safety:** Agent evaluation & observability (Arize, Phoenix), guardrails, PII redaction, prompt-injection defense (OWASP LLM Top 10)
 - **Models & training:** LLMs (OpenAI, Anthropic, Gemini, Qwen, Llama, Mistral), fine-tuning (LoRA/QLoRA), distillation into small models, embeddings, encoder models (BERT, DeBERTa), Whisper
 - **Inference:** vLLM, llama.cpp/GGUF, ONNX for fast CPU inference, constrained/structured decoding (grammars, JSON schema)
- **Calibration & uncertainty:** calibrated model confidence (temperature scaling, Brier, ECE), conformal prediction, "not enough evidence" detection, so a model can hand over instead of guessing
 - **Extraction & retrieval:** multilingual NER and span tagging, cause/effect relation extraction, GLiNER, cross-encoder rerankers, knowledge graphs and multi-hop retrieval
 - **Benchmarking:** reproducible evaluation sets in several languages, noise floors between runs, and reporting the experiments that failed
 - **Voice & data:** Voice agents (Retell), synthetic data generation (NER), NLP & recommendation, scikit-learn, pandas, PyTorch, TensorFlow

Languages	Data & Vector Search	Infrastructure & Architecture
▪ Python, Go, TypeScript, SQL, Scala, Java	▪ Vector: pgvector, FAISS, Annoy (contributor), Pinecone, Qdrant ▪ Data: ETL/ELT, streaming, feature stores, warehousing, analytics	▪ Kubernetes, Docker, Terraform (IaC), CI/CD (GitHub Actions), microservices, distributed systems, AWS, GCP

Education

MSc coursework - Electronics & Electrical Engineering (Machine Learning focus)

2010 - 2012

Bilkent University, Ankara

Focus: Machine Learning applied to signal/image/audio processing. Developed classification models using SVMs, including full ML pipeline: data collection, feature engineering, model development, testing, and validation. Technologies: MATLAB, Python, Support Vector Machines. Worked as Research and Teaching Assistant.

Bachelor of Science - Electronics & Electrical Engineering

2006 - 2010

Bilkent University, Ankara

Spoken Languages

- **Turkish:** Native
- **English:** Full Professional Proficiency